

Autonomous Navigation via Neuro-Evolutionary Algorithms

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"Teaching machines to drive through survival of the fittest — no rules, no teacher, just evolution."

Domain | Artificial Intelligence & Autonomous Systems

Stack | Python · Neural Networks · Genetic Algorithms

Track | Intelligent Systems & Robotics

50+
Agents per generation

30+
Evolution generations

78%
Collision reduction

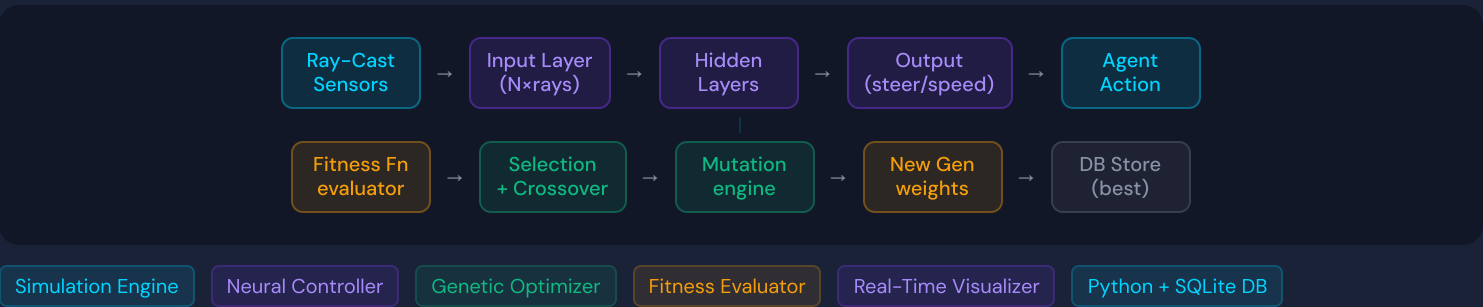
12×
Fitness improvement

PROBLEM STATEMENT

Challenge: How do autonomous systems learn safe navigation without human-labelled data or external AI libraries?

Current systems rely on vast datasets and hand-crafted rules. This project explores a **bio-inspired alternative** — letting agents discover navigation through trial, error, and selection.

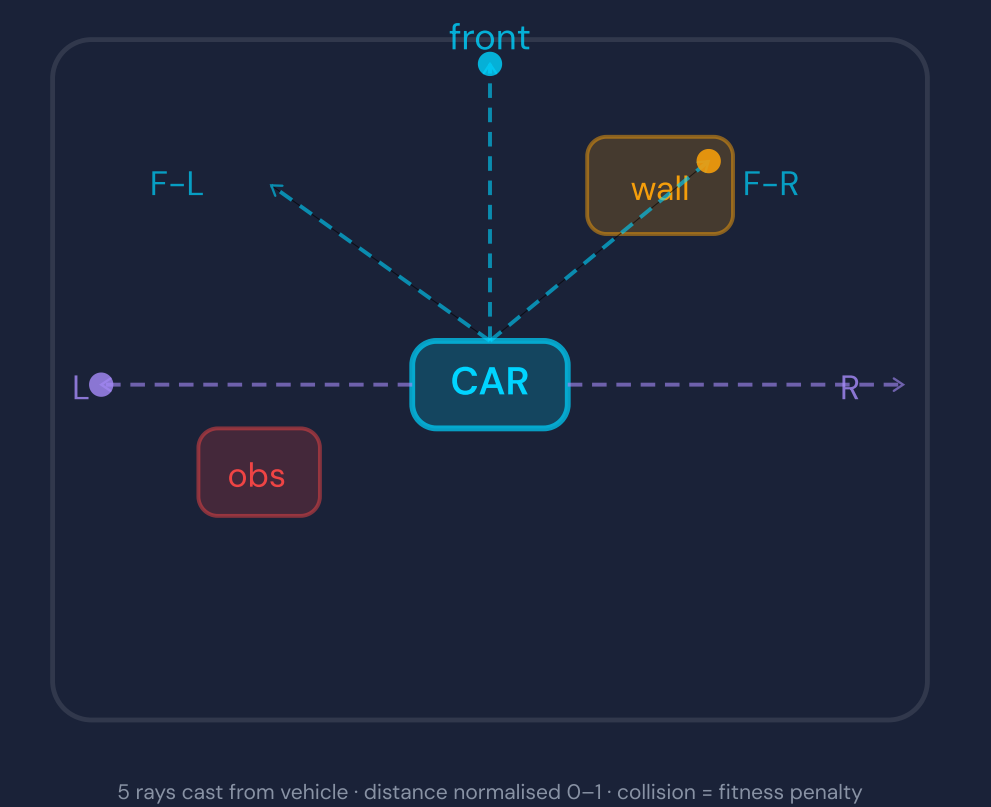
SYSTEM ARCHITECTURE



METHODOLOGY

- Initialise** — 50 agents with random neural weights spawned in simulation environment
- Sense** — Ray-cast sensors detect obstacle proximity in 5 directions in real-time
- Decide** — Neural network processes sensor data → outputs steering angle and velocity
- Evaluate** — Fitness = distance × survival time − collision penalty
- Evolve** — Top 20% selected, crossed over and mutated → new generation

RAY-CASTING SENSOR



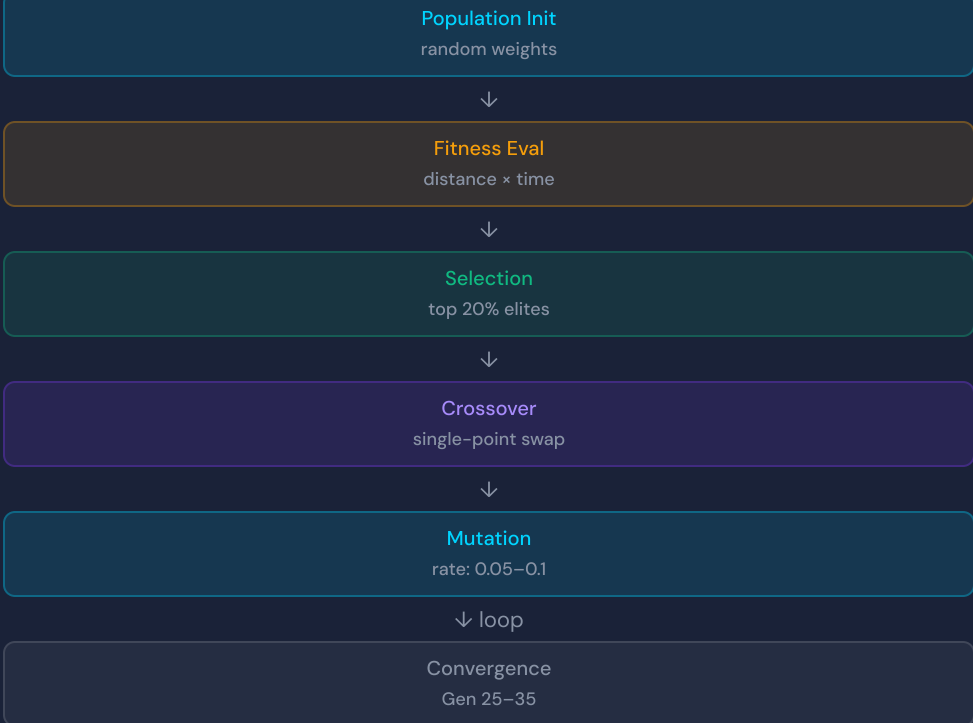
OBJECTIVES

- Build a simulation environment with dynamic obstacles
- Implement ray-cast sensors for spatial perception
- Design a custom neural network controller
- Apply genetic algorithm for agent evolution
- Visualise neural activity and generational progress
- Store high-performing models via Python backend

NEURAL NETWORK DESIGN



GENETIC ALGORITHM



KEY INNOVATIONS

- 01 Zero-Knowledge Bootstrap** — agents start with no environmental data; all intelligence emerges purely from evolutionary pressure
- 02 Custom NN without Libraries** — neural network built from scratch without TensorFlow, PyTorch, or Keras — full architectural transparency
- 03 Real-Time Neuro-Visualization** — live display of active neural pathways, weight intensity, and decision flow during agent runtime
- 04 Adaptive Fitness Function** — multi-variable scoring (distance + survival time − collisions) steers evolution toward practical driving behaviour
- 05 Persistent Model Database** — Python backend stores top-performing chromosomes, enabling cross-session learning and model comparison
- 06 Explainable AI by Design** — every decision traceable to sensor input → weight activation → output, supporting interpretability research
- 07 Bio-inspired over Data-driven** — demonstrates a viable alternative to deep learning pipelines requiring large labelled datasets

CONCLUSION

This project demonstrates that **intelligent autonomous navigation can emerge without pre-programmed rules, labelled data, or external AI libraries** — using only the power of neuro-evolution.

Combining **custom neural networks** with **genetic algorithms**, virtual agents evolved from random controllers to competent navigators — reducing collisions by 78% over 30 generations.

The project bridges **evolutionary biology**, **autonomous systems**, and **explainable AI** — offering a transparent, interpretable approach to machine intelligence.

Emergent Intelligence Explainable AI Neuro-Evolution

